

QP No. MECH-001	Issue Date 15-09-2026	Revision No. A	Date 15-09-2026	Name A. Kulkarni
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MECHANICAL DRIVE SHAFT ENGINEERING SPECIFICATION

Document Code: MECH-001 Discipline: Mechanical Classification: Training / Test

1. Scope

This engineering specification defines requirements for precision machining, dimensional inspection and documentation of a centrifugal pump drive shaft.

2. Material Requirements

Shaft material shall be alloy steel suitable for continuous industrial service. Material certificates shall be traceable to the heat number.

3. Dimensional Requirements

Parameter	Nominal	Tolerance	Unit	Inspection Method
Overall Length	850.00	±0.50	mm	Vernier / CMM
Journal Diameter	75.00	±0.03	mm	Micrometer
Keyway Width	18.00	±0.05	mm	Gauge
Runout	0.00	≤0.05	mm	Dial Indicator

4. Surface Finish

Journal surface finish shall be Ra 1.6 µm maximum. The shaft shall be free from visible burrs, cracks and machining damage.

5. Operating Conditions

Maximum operating temperature: 80.0 °C. Maximum operating pressure: 10 bar. Continuous rated speed: 1480 rpm.

6. Inspection and Records

Inspection results shall identify the measured value, unit, instrument identification, calibration status and inspector. Any out-of-tolerance result shall be recorded as a nonconformance.

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SIGNATURE & APPROVAL RECORD

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Document control statement: This page records preparation, checking and approval of the controlled training/test document.